Tutorial 5B

Analysis of LTI System in S-domain

1. Find the inverse Laplace transform of

$$X(s) = \frac{5s + 13}{s(s^2 + 4s + 13)}, Re(s) > 0$$

2. Find the inverse Laplace transform of

$$X(s) = \frac{s^2 + 2s + 5}{(s + 3)(s + 5)^2}, Re(s) > -3$$

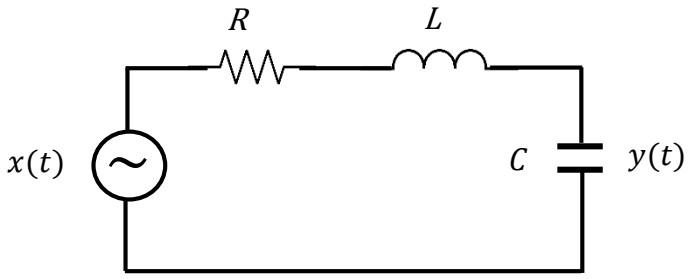


Figure 1

3. Figure 1, shows the series RLC circuit with input voltage $x(t)$ and output voltage $y(t)$. The input-output relation is described in differential equation

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y(t) = x(t),$$

with the initial condition of $y(0^-) = y'(0^-) = 1$ and the input $x(t) = u(t)$. By using Laplace transform approach,

a) determine the overall output response, $y(t)$.

b) determine the transfer function, $H(s)$.

c) if the input signal is $x(t) = 5e^{-0.1t}u(t)$, determine the output response $y(t)$.